AgriDoctor: A Multimodal Intelligent Assistant for Agriculture

Research Paper Explainer

Speed Read

The research paper AgriDoctor: A Multimodal Intelligent Assistant for Agriculture authored by Zhang et al. (2025), focuses on providing an interactive multimodal (text + image) system to help farmers with crop disease detection and general agricultural assistance. AgriDoctor is an AI assistant that helps farmers diagnose crop diseases by combining images, expert knowledge, and conversational guidance in one system. The primary contribution is its modular multimodal framework that integrates expert knowledge retrieval with visual diagnosis, enabling more accurate and context-aware decision-making for agricultural tasks. AgriDoctor is designed for farmers, plant pathologists, and agricultural researchers who need real-time and explainable diagnostic support to address crop diseases and optimize decision-making. What makes AgriDoctor different from typical AI models is that it doesn't rely on a single model to do everything. Instead, it acts like a system of specialized AI tools working together, combining vision, retrieval, and reasoning to produce more reliable results.

The foundation of global food security and sustainable farming relies on accurate and timely diagnosis of crop diseases. Traditionally, artificial intelligence in this field has focused on unimodal techniques such as image-based classification or object detection. While these methods can recognize visual patterns, they lack the ability to model deep agricultural expertise or engage in interactive natural language-based dialogue with farmers. While general Large Language Models (LLMs) and Large Vision-Language Models (LVLMs) have advanced rapidly, they still struggle in agricultural contexts because they lack domain-specific adaptation and hallucinate when faced with specialized terms.

The limitation of current AI systems to integrate expert knowledge with visual evidence restricts their real-world benefits. To achieve truly sustainable agriculture, farmers need tools that go beyond disease identification and provide more insightful information through a conversational interface. The conversational aspect of such an interface can allow farmers to go over all the details with the AI and make reasonable decisions. AgriDoctor builds upon recent advancements in LVLMs such as GPT-4o and Qwen2.5-VL, which achieve success in open-domain tasks but lack agricultural depth.

What makes AgriDoctor different from typical AI models is that it doesn't rely on a single model to do everything. Instead, it uses a system of specialized AI components (handling vision, knowledge retrieval, and reasoning separately), before combining them into a final answer. This framework moves beyond being a general model by integrating five core components. Router, which is for classifying intent and identifying the task at hand. Classifier, which is for high-level visual categorization of crop diseases. Detector, which is for precise spatial localization of symptoms. Retriever, which is for collecting domain-specific agricultural knowledge relevant to a user's textual input. Retriever provides expert context for the task at hand. Finally, an LLM which is for putting together outputs of other components and providing a final context-aware response. The system takes in both images and text, processes them through these components, and generates a clear, actionable response for the user. To create the knowledge base for the retriever and address the data scarcity highlighted in recent literature, the researchers also introduce AgriMM which is a massive new benchmark containing 400,000 annotated images and 831 expert-curated knowledge entries.

Methodology

AgriDoctor decomposes agricultural diagnosis into five coordinated components: Identifying the task at hand (router), classifying the disease type (classifier), locating disease symptoms (detector), retrieving expert knowledge (retriever), and generating the final response (LLM). This modular agent-based multimodal reasoning approach is designed to handle complex agricultural diagnostic tasks more effectively than traditional unimodal systems. Unlike those systems, AgriDoctor separates visual recognition, spatial localization, and knowledge retrieval using a central reasoning engine to combine the outputs into a natural language response.

The framework relies on a specialized dataset, AgriMM, for fine-tuning its individual components to ensure domain-specific accuracy. When a user provides a query, the system first passes it through the router. This component performs language detection (supporting Chinese and English) and intent classification to determine which specialized tool is required. It identifies whether the user wants to classify a disease, locate lesions, or ask a general knowledge-based question. Based on the router's decision, in the next component, the input is forwarded one of three primary tools which are classifier, detector, and retriever. The classifier identifies the specific disease type from an image. It uses a weighted cross-entropy loss during training to handle long-tailed data distributions, ensuring that even rare diseases are accurately recognized. Detector performs high-precision spatial localization, describing exactly where the symptoms are on the crop using bounding boxes. Retriever summarizes the textual query using keyword extraction and searches a curated database of 831 expert entries to find relevant information on disease transmission, impacts, and control strategies. In the last component, the final step involves an LLM-based fusion process. The selected tool's output, for example a disease label or a knowledge snippet, is integrated into a structured prompt alongside the original user query. The LLM then utilizes all this information to generate a coherent, context-aware, and interpretable response. You can think of AgriDoctor like a team of specialists: one identifies the task, another analyzes the image, another retrieves expert knowledge, and a final system combines everything into a useful response.

The framework utilizes a number of state-of-the-art models, each fine-tuned for the agricultural domain. BERT, Bidirectional Encoder Representations from Transformers, is used in the router for semantic understanding and intent inference. It is also used in the retriever to embed queries into dense vectors for similarity searching. CLIP, Contrastive Language-Image Pre-training, with ViT, Vision Transformer, serves as the backbone for the classifier. The researchers kept the visual encoder fixed while fine-tuning a trainable classification head on 138 disease categories. YOLOv12, You Only Look Once, a single stage object detection architecture, is used in the detector. This model was chosen for its ability to handle multi-scale feature aggregation and transformer-based attention, which is critical for identifying small or obstructed lesions. FAISS, Facebook AI Similarity Search, is utilized by the retriever to perform fast similarity metrics over high-dimensional vector spaces, ensuring accurate retrieval of expert knowledge. Finally, for the LLM component, the study utilized models like DeepSeek-V3 for automated evaluation and Qwen2.5 variants for benchmarking performance.

The effectiveness of these models is attributed to the AgriMM benchmark which is a massive dataset constructed specifically for this methodology. It includes 400,000 annotated images covering 29 crops and 138 diseases, a bilingual expert knowledge base, and 300,000 prompt examples used to train the router for intent-focused tool selection.

General workflow of AgriDoctor and preparation process for AgriMM is provided in the figure below. The AgriDoctor on the left shows input, router, classifier, detector, retriever, and LLM respectively. The flow as described before is evident in the figure. The AgriMM preparation process from raw data to ready-to-use data for the retriever is also shown on the right side. The figure highlights modular multimodal structure of AgriDoctor where the user queries are routed through disease classification, lesion detection, or knowledge retrieval tools followed by LLM-based output fusion. The data construction workflow includes image annotation and knowledge structuring from raw visual and textual sources. This workflow provides valuable information which supports the training of AgriDoctor's modules. (Zhang et al., 2025)

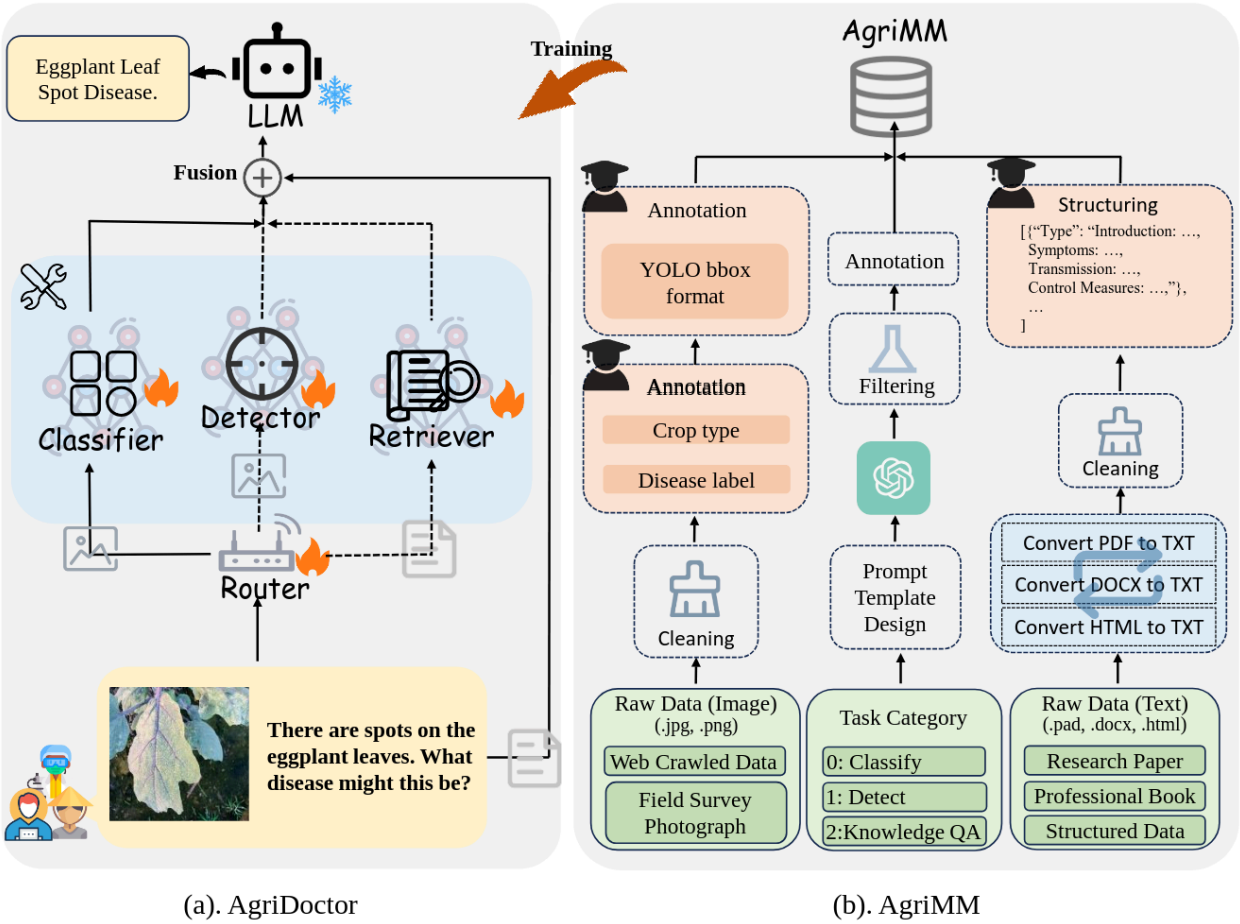

Fig 1: (a) Overall architecture of AgriDoctor, (b) Data construction workflow of AgriMM (Zhang et al., 2025)

The researchers conduct testing across three categories of disease classification, lesion detection, and knowledge-grounded question answering. To ensure an unbiased assessment, the researchers utilized a balanced test set of 300 samples that were excluded from training. The evaluation metrics are based on two principles of semantic consistency and information completeness. Semantic consistency measures the semantic alignment between the model output and the expert-validated reference answers. It also reflects the accuracy of the model in classifying visual categories. Information completeness evaluates the extent to which the model output covers the disease categories and diagnostic evidence. DeepSeek-V3 model is utilized for evaluation based on the two principles. The input and output of AgriDoctor, the reference answer, and the evaluation criteria is given to DeepSeek-V3 and a score between 0 to 1 is produced as an evaluation metric. While using DeepSeek-V3 as an automatic evaluator offers a convenient and scalable assessment

mechanism, it may introduce potential biases or hallucinations inherent to language models. To mitigate such risks, the researchers additionally performed human verification on a subset of the evaluation results to ensure the reliability and validity of the scoring.

AgriDoctor achieves significantly higher performance than general AI models, meaning it produces more accurate diagnoses and more complete, useful explanations. AgriDoctor achieves an overall performance score of 0.8630, consistently surpassing state-of-the-art models like GPT-4o-mini (0.7043) and Qwen2.5-VL (0.6593). The classification tool reached an accuracy of 96.2% on the test set, proving that pre-trained visual features from CLIP can be effectively adapted to plant pathology. In comparative evaluations using DeepSeek-V3 as the judge, AgriDoctor maintained better semantic consistency and information completeness compared to general LVLMs. The detection tool which utilizes the YOLOv12 architecture achieves 0.876 precision and 0.893 recall. This indicates a strong ability to both locate and correctly identify diseased regions even when lesions are small or obstructed. The retrieval tool is the framework's strongest area, achieving a semantic consistency score of 0.965 and information completeness score of 0.988. The model successfully retrieved and combined expert content for queries regarding disease transmission mechanisms and control strategies, significantly outperforming a fine-tuned version of Qwen2.5-14B-Instruct. The BERT-based router demonstrated near-perfect intent recognition, with an accuracy of 99.00% for English and 98.82% for Chinese tasks. Its lightweight design ensures that the system remains practical for real-time use by minimizing inference costs. The prompts for DeepSeek-V3 evaluation are shown below for each component. This figure highlights how textual prompts are used to describe the measures for each evaluation task. (Zhang et al., 2025)

"**{text}** is the input text, **{reference_answer}** is the reference answer, and **{agri_agent}** is the output generated by the model.
Please evaluate the model output based on the following two criteria:

1. Semantic Consistency: If the predicted disease category matches the reference answer, it is considered semantically consistent and receives a full score of 1. If the output includes multiple categories and none of them match the reference, the score is 0. If at least one of the predicted categories matches the reference, assign a score of 0.5.

2. Information Completeness: Evaluate whether the output contains essential information such as the disease category. If the disease category is missing or labeled as 'unknown', assign a score of 0.

Each dimension is scored on a scale of 0 to 1.
The final score is calculated as the average of the two dimensions.
Please report the individual scores for semantic consistency, information completeness, and the final averaged score."

Classification evaluation criteria

"**{text}** is the input text, **{reference_answer}** is the reference answer, and **{agri_agent}** is the output generated by the model.
Please evaluate the model output based on the following two criteria:

1. Semantic Consistency: If the predicted disease category matches the reference answer, the output is considered semantically consistent and should receive a full score of 1. If the disease category is incorrect or not provided, the score should be 0.

2. Information Completeness: The output should include two key elements: the disease category and the detection region(s). If detection regions are provided, please check the number of regions and assess the similarity of their values compared to the reference. These two components reflect the completeness of the information conveyed.

Each dimension is scored on a scale from 0 to 1.
The final score is the average of the two dimensions.
Please report the scores for semantic consistency, information completeness, and the final overall score."

Detection evaluation criteria

"**{text}**" is the input text, "**{reference_answer}**" is the reference answer, and "**{agri_agent}**" is the model output.
Please evaluate the model output based on the following two criteria:

1. Semantic Consistency: Whether the model output is semantically similar to the reference answer.

2. Information Completeness: The extent to which the output contains complete and relevant information.

Each criterion is scored on a scale from 0 to 1.
The final score is the average of the two criteria.
Please report the individual scores for semantic consistency, information completeness, and the final overall score.

Retrieval evaluation criteria

Fig 2: Evaluation criteria for Deepseek-V3 targeting classifier, detector, and retriever (Zhang et al., 2025)

Takeaways

This research highlights a broader shift in AI: instead of relying on a single all-purpose model, better results are achieved from combining multiple specialized systems that work together. By leveraging the AgriMM dataset, the framework achieves high-precision diagnostics and provides expert knowledge interaction.

Together, AgriDoctor and the AgriMM dataset show how combining structured knowledge with AI can significantly improve decision-making in complex, real-world environments. General LVLMs struggle with fine-grained agricultural tasks due to a lack of specialized visual concepts and expert knowledge. AgriDoctor proves that domain-specific fine-tuning and structured knowledge integration are essential for reliable performance. By separating the classifier and detector from the LLM reasoning engine, the system achieves higher interpretability. This design allows for the integration of new tools or modalities without requiring a total redesign of the central model. The AgriMM dataset fills a significant gap in the field, providing the expert information required for fine-grained plant pathology which can be useful for future research. By using a knowledge retriever to base responses on verified expert entries, AgriDoctor mitigates hallucination present in general LLMs.

While highly effective, the researchers also note that the system currently has limited coverage of rare crop species and uncommon disease variants. The bilingual support is robust but expanding into low-resource languages is necessary for global deployment in diverse agricultural regions. Future research will likely focus on making the assistant more adaptive to regional conditions and a broader array of plant species.

Why This Matters at Work

For professionals building AI tools in specialized industries, this research shows that accuracy doesn't just come from better models, it comes from better system design. By creating and releasing AgriMM the authors provide a foundational resource that addresses the scarcity of high-quality domain-annotated data which has historically hindered progress in specialized AI. This can help other researchers in their future work. For plant pathologists, AgriDoctor offers a practical real-time tool for interactive diagnosis. Unlike traditional models, AgriDoctor's modular design allows professionals to receive not just a diagnosis, but also precise visual localizations and expert advice on plant disease transmission and control strategies. The use of a lightweight BERT-based router also showcases that performant AI can be made computationally efficient for real-time deployment in industries where inference cost and speed are important.

To ensure the validity of the automated evaluations, the researchers performed human verification on a subset of the results to mitigate biases or errors inherent in the AI judges used during testing. While powerful, the system is designed to support, not replace, human expertise, and still has limitations in coverage and language support. It still relies on human experts to make the decisions. It is important to note that the model is still not completely reliable and robust. It also provides support for English and Chinese which limits global deployment.

The researchers mention several directions for future exploration to make agricultural AI more inclusive and robust. Future research needs to address the lack of data for rare crop species, uncommon disease variants, and region-specific conditions that were not fully captured in the current benchmark. While the system currently supports English and Chinese, expanding multilingual capabilities is essential to make the tool accessible to farmers in low-resource language areas. Developing a more globally deployable and adaptive assistant that can adjust to different climates and regional agricultural practices remains a primary goal for the next generation of this technology.

While this system is designed for agriculture, the same approach could be applied in fields like healthcare, energy, or environmental science, where decisions depend on both visual data and expert knowledge.

References

Zhang, M., Xu, Z., Wang, P., Li, R., Wang, L., Liu, Q., ... & Wu, S. (2025). AgriDoctor: A Multimodal Intelligent Assistant for Agriculture. *arXiv preprint arXiv:2509.17044*.

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APRIL 2026